The documentation and process conversion measures necessary to comply with this revision shall be completed by 20 August 1998

INCH-POUND

MIL-PRF-19500/542F
20 April 1998
SUPERSEDING
MIL-S-19500/542E
17 August 1994

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, FIELD EFFECT, N-CHANNEL,
SILICON TYPES 2N6756, 2N6758, 2N6760, 2N6762,
JAN, JANTX, JANTXV, JANS, JANHC, and JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for N-channel, enhancement-mode, MOSFET, power transistors. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500 and two levels of product assurance are provided for each unencapsulated device type.

1.2 Physical dimensions. See figure 1 (TO-204AA; formerly TO-3), and figures 2, 3, for JANHC and JANKC die dimensions. See 6.5 for unencapsulated device types.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Type	P_T 1/ T_C = +25°C	P_T T_C = +25°C (free air)	V_{DS}	V_{DG}	V_{GS}	I_{D1} 2/ T_C = +25°C	I_{D2} 2/ T_C = +100°C
	W	W	V dc	V dc	V dc	A dc	A dc
2N6756	75	4	100	100	± 20	14.0	9.0
2N6758	75	4	200	200	± 20	9.0	6.0
2N6760	75	4	400	400	± 20	5.5	3.5
2N6762	75	4	500	500	± 20	4.5	3.0

Type	I_S	I_{DM} 3/	T_J and T_{STG}	V_{ISO} 70,000 feet altitude	Max $r_{DS(on)}$ 1/ V_{GS} = 10 V dc, $I_D = I_{D2}$		$R_{\theta JC}$ max
					T_J = 25°C	T_J = 150°C	
	A dc	A (pk)	°C		ohms	ohms	°C/W
2N6756	14.0	56	-55 to +150		0.18	0.36	1.67
2N6758	9.0	36	-55 to +150		0.4	0.84	1.67
2N6760	5.5	22	-55 to +150	400	1.0	2.5	1.67
2N6762	4.5	18	-55 to +150	500	1.5	3.75	1.67

1/ Derate linearly 0.6 W/°C for T_C > +25°C. $P_T = \dfrac{T_{J\ max} - T_C}{R_{\Theta JC}}$

2/ $\sqrt{\dfrac{T_J\ \max - T_C}{\left(R_{\Theta JC}\ x\ R_{DS(on)}\ at\ T_J\ \max\right)}}$

3/ I_{DM} = 4 I_{D1}; I_{D1} as calculated in note 2.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Primary electrical characteristics. Unless otherwise specified, T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 V I_D = 1 mA dc	$V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ I_D = 0.25 mA		Max I_{DSS1} V_{GS} = 0 V	Max $r_{DS(on)1}$ 1/ V_{GS} = 10 V dc $I_D = I_{D2}$
				V_{DS} = 80 percent of rated V_{DS}	T_J = +25°C
	V dc	V dc		µA dc	ohms
		Min	Max		
2N6756	100	2.0	4.0	25	0.18
2N6758	200	2.0	4.0	25	0.4
2N6760	400	2.0	4.0	25	1.0
2N6762	500	2.0	4.0	25	1.5

1/ Pulsed, (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

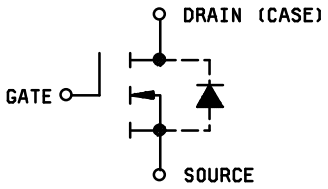
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

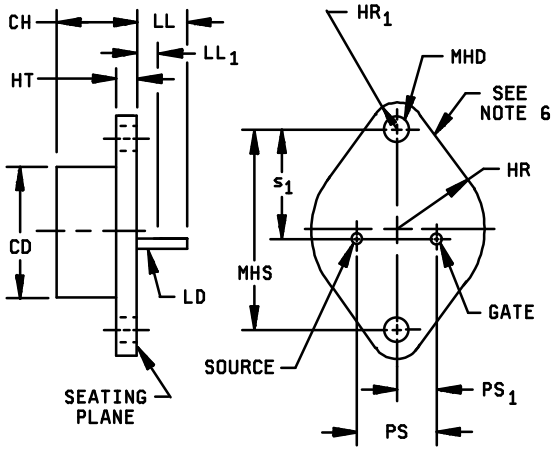
2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD		.875		22.23	
CH	.250	.360	6.35	9.14	
HR	.495	.525	12.57	13.34	
HR_1	.131	.188	3.33	4.78	
HT	.060	.135	1.52	3.43	
LD	.038	.043	0.97	1.09	
LL	.312	.500	7.92	12.70	
LL_1		.050		1.27	
MHD	.151	.161	3.84	4.09	
MHS	1.177	1.197	29.90	30.40	
PS	.420	.440	10.67	11.18	3, 5
PS_1	.205	.225	5.21	5.72	3, 5
s_1	.655	.675	16.64	17.15	



CIRCUIT SCHEMATIC

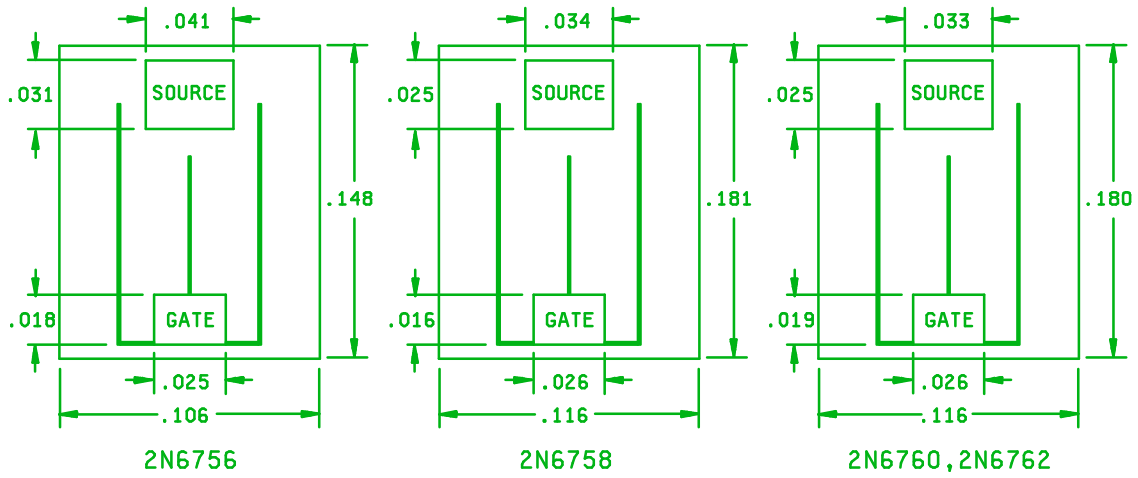


NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. These dimensions should be measured at points .050 inch (1.27 mm) and .055 inch (1.40 mm) below seating plane. When gauge is not used measurement will be made at the seating plane.
4. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
5. Mounting holes shall be deburred on the seating plane side.
6. Drain is electrically connected to the case.

FIGURE 1. Physical dimensions of transistor (TO-204AA).

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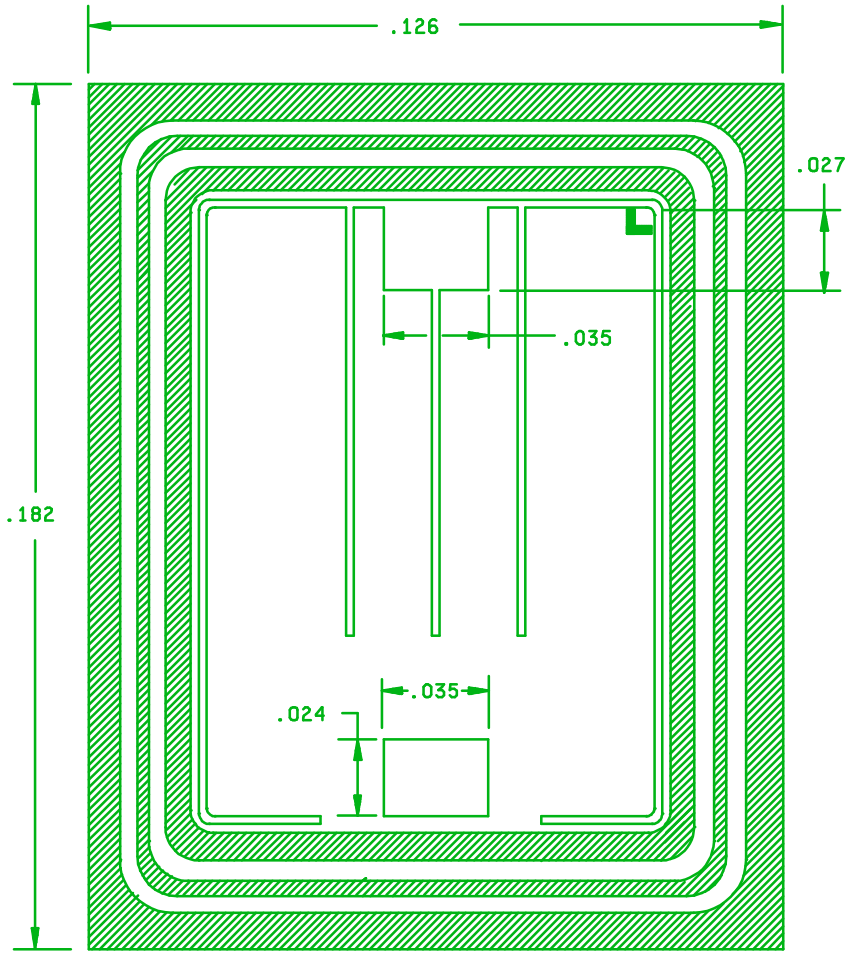


Inch	mm	Inch	mm	Inch	mm
.016	0.41	.031	0.79	.116	2.95
.018	0.46	.033	0.84	.148	3.76
.019	0.48	.034	0.86	.180	4.57
.025	0.64	0.41	1.04	.181	4.60
.026	0.66	.106	2.69		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ".005 inch (0.13 mm).
4. The physical characteristics of the die thickness are .0187 inch (0.475 mm). The back metals are chromium, nickel and silver. The top metal is aluminum and the back contact is the drain.

FIGURE 2. JANHCA and JANKCA (A-version).

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Inch	mm
.024	0.61
.027	0.69
.035	0.89
.126	3.20
.182	4.62

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ".005 inch (0.13 mm).
4. The physical characteristics are the die thickness .014 inch (0.36 mm). The back metals are aluminum, nickel, and titanium. The top metal is aluminum and the back contact is the drain.

FIGURE 3. JANHCB and JANKCB (B-version) die dimensions for 2N6756, 2N6758, 2N6760, and 2N6762.

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3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.3 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 (TO - 204AA) herein.

3.3.1 Internal construction. Multiple chip construction is not permitted to meet the requirements of this specification.

3.3.2 Lead finish. Lead finish shall be solderable as defined in MIL-STD-750, MIL-PRF-19500, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection.

3.5.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of electrostatic charge. The following handling practices shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100$ k, whenever bias voltage is to be applied drain to source.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3 herein.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-PRF-19500 and table II herein.

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4.3 Screening (JANS, JANTX and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (Appendix E, table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/	Gate stress test (see 4.5.5)	Gate stress test (see 4.5.5)
1/ 2/	Method 3470 (see 4.5.4) optional	Method 3470 (see 4.5.4) optional
1/	Method 3161 (see 4.5.3)	Method 3161 (see 4.5.3)
9	I_{GSS1}, I_{DSS1}	
10	Method 1042, test condition B	Method 1042, test condition B
11	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$ Subgroup 2 of table I herein: ΔI_{GSS1} = ± 20 nA dc or ± 100 percent of initial value, whichever is greater. ΔI_{DSS1} = ± 25 µA dc or ± 100 percent of initial value, whichever is greater.	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$, subgroup 2 of table Iherein.
12	Method 1042, test condition A, t = 240 hours	Method 1042, test condition A; or t = 48 hours minimum at +175°C min
13	Subgroups 2 and 3 of table I herein: ΔI_{GSS1} = ± 20 nA dc or ± 100 percent of initial value, whichever is greater. ΔI_{DSS1} = ± 25 µA dc or ± 100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ± 20 percent of initial value. $\Delta V_{GS(th)1}$ = ± 20 percent of initial value.	Subgroup 2 of table I herein: ΔI_{GSS1} = ± 20 nA dc or ± 100 percent of initial value, whichever is greater. ΔI_{DSS1} = ± 25 µA dc or ± 100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ± 20 percent of initial value. $\Delta V_{GS(th)1}$ = ± 20 percent of initial value.

1/ Shall be performed anytime before screen 10.
2/ This test method in no way implies a repetitive avalanche energy rating. This test need not be performed in group A when performed as a screen.

4.3.1 Screening (JANHC and JANKC). Screening of die shall be in accordance with appendix E, table IV of MIL-PRF-19500. As a minimum Die, shall be 100 percent probed per group A, subgroup 2 except test current shall not exceed 20 amperes.

4.3.2 JANHC and JANKC die. Qualification shall be in accordance with appendix E, table IV of MIL-PRF-19500.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. Alternate flow is allowed for conformance inspection in accordance with appendix E of MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V of MIL-PRF-19500 and table I herein. Electrical measurements (end-points) shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-PRF-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, appendix E, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	1051	Test condition G.
B4	1042	Test condition D; the heating cycle shall be 1 minute minimum, 2,000 cycles. No heat sink nor forced air cooling on the device shall be permitted during the cycle.
B5	1042	Accelerated steady-state operation life; test condition A, V_{DS} = rated T_A = +175°C, t = 120 hours minimum. Read and record $V_{(BR)DSS}$ (pre and post at 1 mA = I_D. Read and record I_{DSS} (pre and post). Deltas for $V_{(BR)DSS}$ shall not exceed 10 percent and I_{DSS} shall not exceed 25 µA. Accelerated steady-state gate stress; condition B, V_{GS} = rated, T_A = +175°C, t = 24 hours.
B5	2037	Bond strength (Al-Au die interconnects only); test condition A.
B6	3161	See 4.5.2.

4.4.2.2 Group B inspection, appendix E, table VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Conditions
B2	1051	Test condition G.
B3	1042	Test condition D, 2,000 cycles minimum. The heating cycle shall be 1 minute minimum.
B3	2037	Test condition A. All internal bond wires for each device shall be pulled separately.
B4	2075	See 3.3.2.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table I, group A, subgroup 2 herein.

4.4.3.1 Group C inspection, appendix E, table VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	1056	Test condition A.
C2	2036	Test condition A; weight = 10 lbs, t = 15 s.
C6	1042	Test condition D; 6,000 cycles minimum. The heating cycle shall be 1 minute minimum.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. $R_{\theta JC}$ max = 1.67°C/W.

a. I_M measuring current ..... 10 mA.

b. I_H drain heating current ..... 1.5 A.

c. t_H heating time ..... Steady-state (see MIL-STD-750, method 3161 for definition).

d. V_H drain-source heating voltage ..... 20 V.

e. t_{MD} measurement time delay ..... 30 to 60 µs

f. t_{SW} sample window time ..... 10 µs (max).

4.5.3 Thermal response (ΔV_{SD} measurements). The ΔV_{SD} measurements shall be performed in accordance with method 3161 of MIL-STD-750. The ΔV_{SD} conditions (I_H and V_H) and maximum limit shall be derived by each vendor from the thermal response curves (see figure 4). The ΔV_{SD} measurement and conditions for each device in the qualification lot shall be submitted (read and record) in the qualification report. The chosen ΔV_{SD} shall be considered final after the manufacturer has had the opportunity to test five consecutive lots. The following parameter measurements shall apply:

a. I_M measuring current ..... 10 mA.

b. I_H drain heating current ..... 1.5 A (min).

c. t_H heating time ..... 100 ms.

d. V_H drain-source heating voltage ..... 20 V (min).

e. t_{MD} measurement time delay ..... 30 to 60 µs.

f. t_{SW} sample window time ..... 10 µs (max).

4.5.4 Single pulse unclamped inductive switching.

a. Peak current, I_D ..... Rated I_D.

b. Peak gate voltage, V_{GS} ..... 10 V.

c. Gate to source resistor, R_{GS} ..... $25 \leq R_{GS} \leq 200$.

d. Initial case temperature ..... +25°C +10, -5°C.

e. Inductance ..... 100 µH minimum.

f. Number of pulses to be applied ..... 1 pulse minimum.

g. Supply voltage V_{DD} ..... 50 V.

4.5.5 Gate stress test.

a. V_{GS} = 30 V minimum.

b. t = 250 µs minimum.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal response 2/	3161	See 4.5.3	$Z_{\theta JC}$		1.6	°C/W
Breakdown voltage, drain to source	3407	V_{GS} = 0 V dc I_D = 1 mA dc, condition C	$V_{(BR)DSS}$			V dc
2N6756				100		
2N6758				200		
2N6760				400		
2N6762				500		
Gate to source voltage (threshold)	3404	$V_{DS} \geq V_{GS}$ I_D = 0.25 mA dc	$V_{GS(th)1}$	2.0	4.0	V dc
Gate current	3411	V_{GS} = ± 20 V dc, V_{DS} = 0 Bias condition	I_{GSS1}		± 100	nA dc
Drain current	3413	V_{GS} = 0 V dc V_{DS} = 80 percent of rated V_{DS}; Bias condition C	I_{DSS1}		25	µA dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, pulsed (see 4.5.1); condition A, I_D = rated I_{D1} (see 1.3)	$r_{DS(on)1}$			ohms
2N6756					0.18	
2N6758					0.4	
2N6760					1.0	
2N6762					1.5	
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, pulsed (see 4.5.1); condition A, I_D = rated I_{D1} (see 1.3)	$r_{DS(on)2}$			ohms
2N6756					.21	
2N6758					.49	
2N6760					1.22	
2N6762					1.80	

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward voltage (source drain diode)	4011	Pulsed (see 4.5.1) V_{GS} = 0 V, I_D = I_{D1}	V_{SD}			V
2N6756					1.8	
2N6758					1.6	
2N6760					1.5	
2N6762					1.4	
Subgroup 3						
High temperature operation:		T_C = T_J = +125°C				
Gate current	3411	Bias condition C V_{GS} = ± 20 V dc V_{DS} = 0 V dc	I_{GSS2}		± 200	nA dc
Drain current	3413	Bias condition C V_{GS} = 0 V dc V_{DS} = 100 percent of rated V_{DS}	I_{DSS2}		1.0	mA dc
		V_{DS} = 80 percent of rated V_{DS}	I_{DSS3}		0.25	mA dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc pulsed (see 4.5.1); I_D = rated I_{D2}	$r_{DS(on)3}$			ohms
2N6756					0.34	
2N6758					0.8	
2N6760					2.2	
2N6762					3.3	
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$ I_D = 0.25 mA dc	$V_{GS(th)2}$	1.0		V dc
Low temperature operation:		T_C = T_J = -55°C				
Gate to source voltage threshold	3403	$V_{DS} \geq V_{GS}$ I_D = 0.25 mA dc	$V_{GS(th)3}$		5.0	V dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Switching time test	3472	I_D = rated I_{D1}; V_{GS} = 10 V dc Gate drive impedance = 7.5Ω V_{DD} = 50 percent of $V_{BR(DSS)}$				
Turn-on delay time			$t_{d(on)}$			ns
2N6756, 2N6758					35	
2N6760, 2N6762					30	
Rise time			t_r			ns
2N6756, 2N6758					80	
2N6760, 2N6762					40	
Turn-off delay time			$t_{d(off)}$			ns
2N6756, 2N6758					60	
2N6760, 2N6762					80	
Fall time			t_f			ns
2N6756					45	
2N6758					40	
2N6760					35	
2N6762					30	
Subgroup 5						
Safe operating area test	3474	See figure 5; V_{DS} = 80 percent of rated $V_{BR(DSS)}$ V_{DS} = 200 V max; t_p = 10 ms				
Electrical measurements		See table I, group A, subgroup 2				
Single pulse unclamped inductive switching	3470	See 4.5.4, c = 0, 116 devices				
Electrical measurements		See table I, group A, subgroup 2				
Subgroup 6						
Not applicable						

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 7						
Gate charge	3471	Condition B				
On-state charge			$Q_{g(on)}$			nC
2N6756					35	
2N6758					39	
2N6760					39	
2N6762					40	
Gate to source charge			Q_{gs}			nC
2N6756					10	
2N6758					5.7	
2N6760, 2N6762					6.0	
Gate to drain charge			Q_{gd}			nC
2N6756					15	
2N6758, 2N6760, 2N6762					20	
Reverse recovery time	3473	di/dt = 100 A/µs $V_{DD} \leq 30$ V $I_D = I_{D1}$	t_{rr}			ns
2N6756					300	
2N6758					500	
2N6760					700	
2N6762					900	

1/ For sampling plan, see MIL-PRF-19500.
2/ This test is required for the following end points (not intended for Screen 13):

JANS - group B, subgroups 3 and 4
JANTX and JANTXV - Group B, subgroups 2 and 3
Group C, subgroup 6
Group E, subgroup 1

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TABLE II. Group E inspection (all quality levels) for qualification only.

Inspection 1/	MIL-STD-750		Qualification and large lot quality conformance inspection 1/
	Method	Conditions	
Subgroup 1			45 devices c = 1
Temperature cycle	1051	Condition G, 500 cycles	
Hermetic seal			
Fine leak Gross leak			
Electrical measurements		See table I, group A, subgroup 2	
Subgroup 2 2/			45 devices c = 1
Steady-state reverse bias	1042	Condition A, 1,000 hours	
Electrical measurements		See table I, group A, subgroup 2	
Steady-state gate bias	4042	Condition B, 1,000 hours	
Electrical measurements		See table I, group A, subgroup 2	
Subgroup 3			
Not applicable			
Subgroup 4			5 devices c = 0
Thermal resistance	3161	$R_{\theta JC}$ = 1.67°C/W max. See 4.5.2	
Subgroup 5			5 devices c = 0
Barometric pressure (reduced) 400 and 500 V only	1001	Test condition C $V_{ISO} = V_{DS}$ $I_{(ISO)}$ = .25 mA (max)	

1/ JANHC and JANKC devices are qualified in accordance with appendix G of MIL-PRF-19500.
2/ A separate sample for each test may be pulled.

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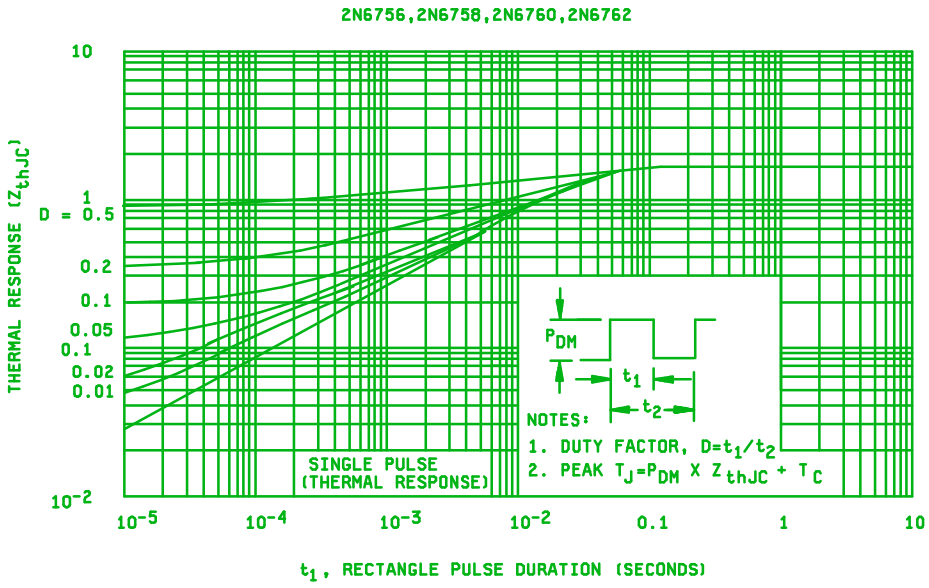


FIGURE 4. Thermal response curves.

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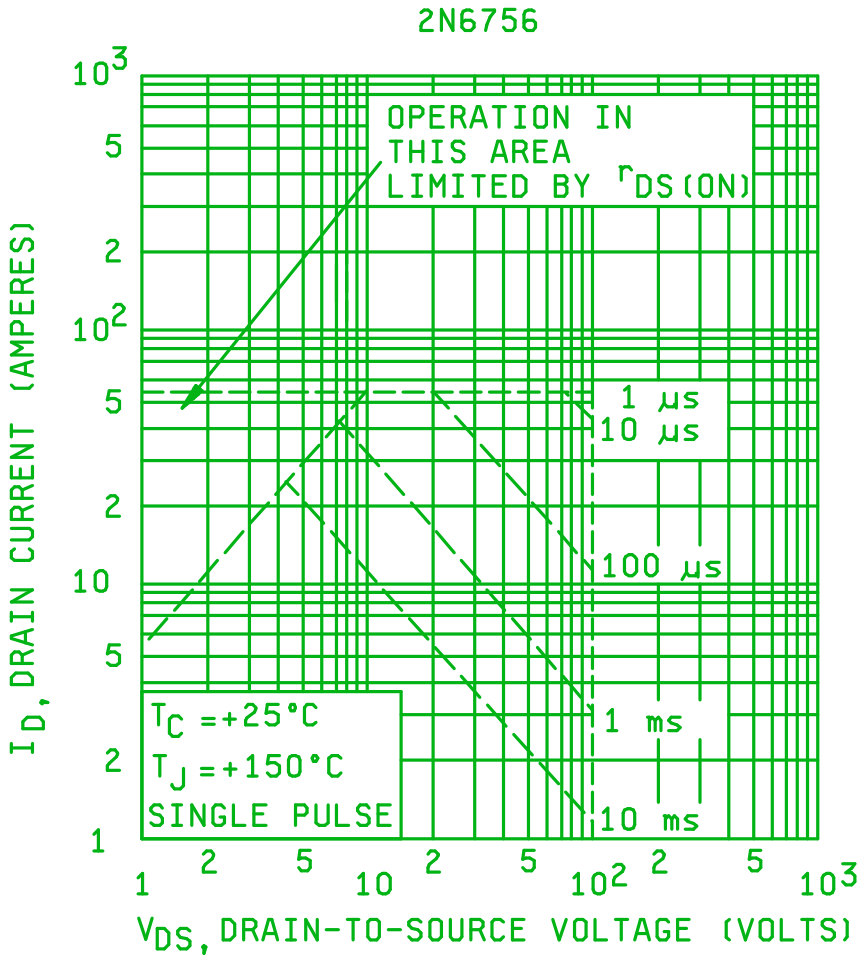


FIGURE 5. Safe operating area.

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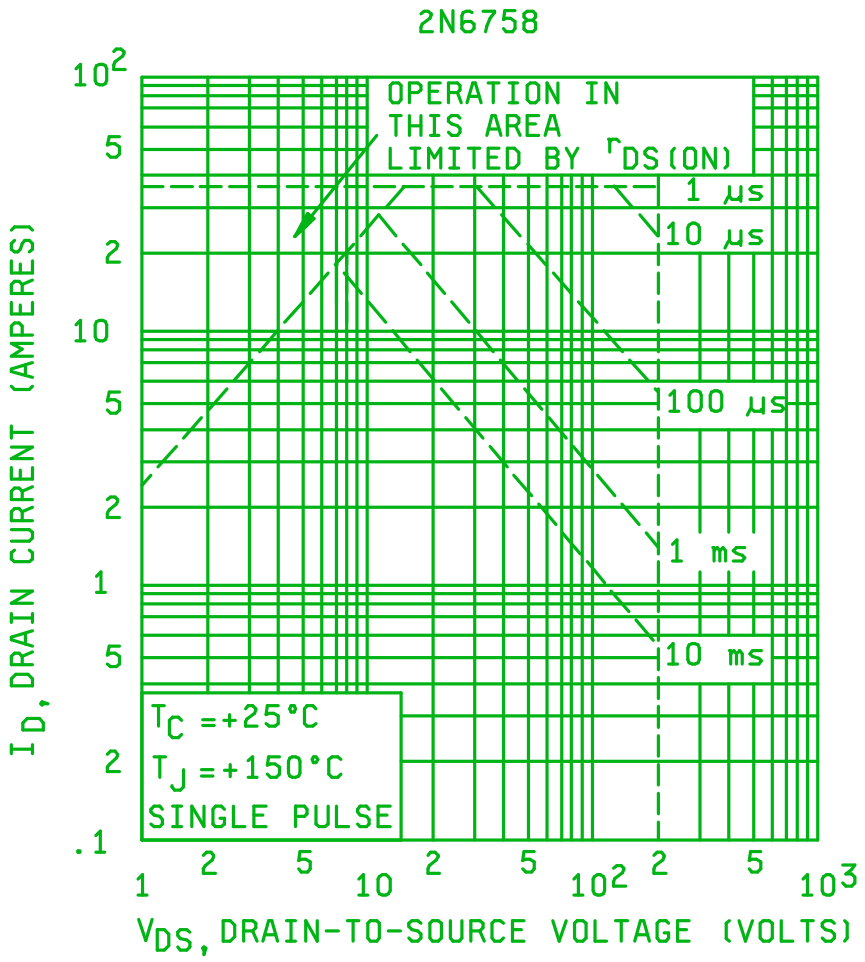


FIGURE 5. Safe operating area - Continued.

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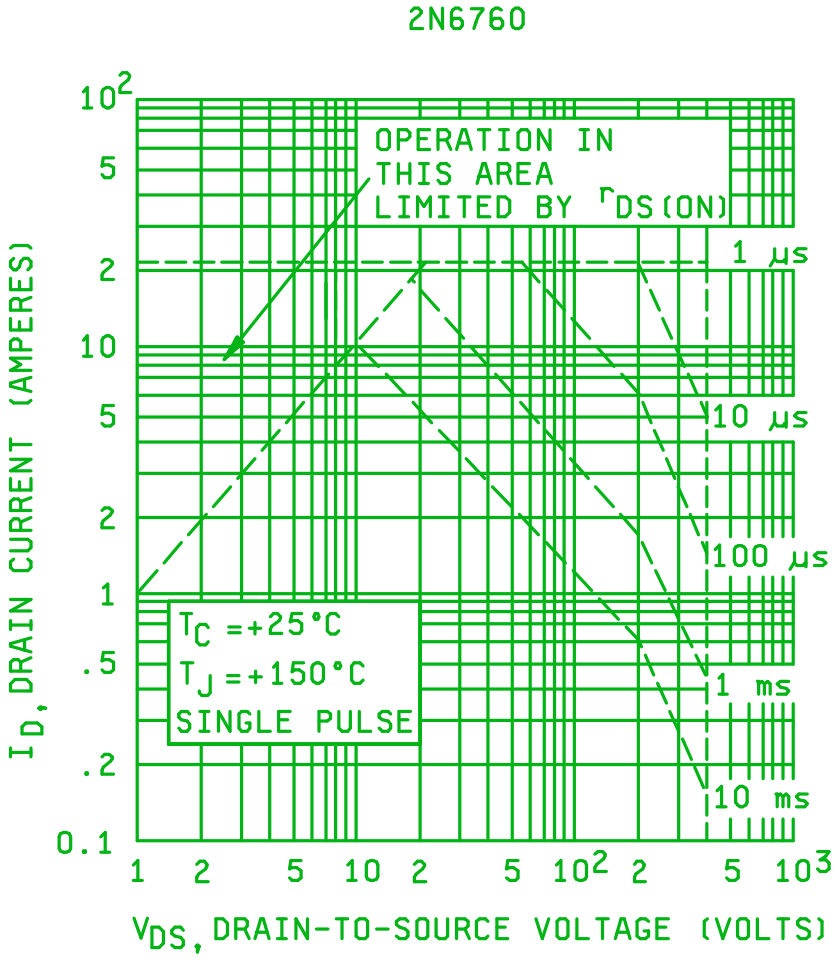


FIGURE 5. Safe operating area - Continued.

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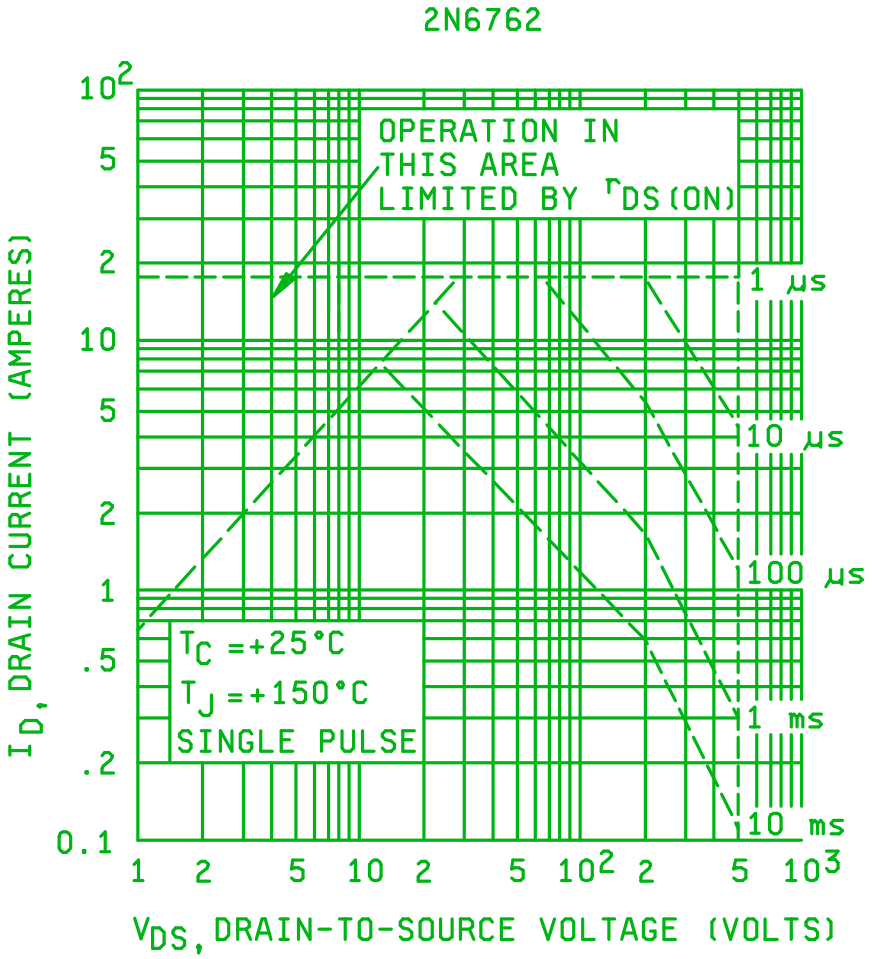


FIGURE 5. Safe operating area - Continued.

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-PRF-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Substitution information. Devices covered by this specification are substitutable for the manufacturers' and users' Part or Identifying Number (PIN). This information in no way implies that manufacturers' PIN's are suitable as a substitute for the military PIN.

PIN	Manufacturer's CAGE code	Manufacturer's and user's PIN
2N6756	17856, 59993, 18722, 04713	IRF130
2N6758	17856, 59993, 18722, 04713	IRF230
2N6760	17856, 59993, 18722, 04713	IRF330
2N6762	17856, 59993, 18722, 04713	IRF430

6.5 Replacement data. JANTX devices shall be a direct replacement for JAN devices (example: JANTX2N6756 for JAN2N6756).

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6.6 Suppliers of JANHC and JANKC die. The qualified die suppliers with the applicable letter version (example JANHCA2N6756) will be identified on the QPL.

JANC ordering information		
PIN	Manufacturers	
	59993	18722
2N6756	JANHCA6756 JANKCA6756	JANHCB6756 JANKCB6756
2B6758	JANHCA6758 JANKCA6758	JANHCB6758 JANKCB6758
2N6760	JANHCA6760 JANKCA6760	JANHCB6760 JANKCB6760
2N6762	JANHCA6762 JANKCA6762	JANHCB6762 JANKCB6762

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Navy - TD
Air Force - 19, 70, 80

Preparing activity:
DLA - CC

(Project 5961-1975)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/542F	**2. DOCUMENT DATE (YYMMDD)** 980420

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTORS, N-CHANNEL, SILICON, TYPES 2N6756, 2N6758, 2N6760, 2N6762 JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. Point of contact: Alan Barone	b. TELEPHONE Commercial 614-692-0510 DSN 850-0510 FAX 614-692-6939 EMAIL alan_barone@dscc.dla.mil
c. ADDRESS: Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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